

Does 4-Bit Quantization Change Multi-Turn Jailbreak Robustness?

A Two-Metric Evaluation of NF4, AWQ, and GPTQ on Llama-3.2-3B and Llama-3.1-8B

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Abstract. Quantization lets large language models run on edge hardware, where real use is multi-turn. Prior work measured quantization’s effect on refusal at a single turn only and reported degradation. This paper gives the first measurement across a multi-turn jailbreak, testing the claim on Llama-3.2-3B (FP16, NF4) and Llama-3.1-8B (FP16, AWQ, GPTQ) with two metrics: a refusal classifier and a Llama Guard 3 harm judge. Quantization does not degrade safety. On standalone AdvBench prompts, single-turn refusal is flat or higher under quantization for both models and all three methods, with no significant comparison. Under CoSafe multi-turn attacks, precision changes neither refusal (3B 23% versus 23%, McNemar $p = 0.62$; 8B 27 to 32%, Cochran Q $p = 0.12$) nor harmful-response rate. The one significant single-turn effect, 8B refusing more after quantization, vanishes under the attack. A safety-prompt positive control confirms the harm metric can detect a real intervention, so the null is a measured absence of effect rather than a lack of power. The evaluation also shows that refusal-based attack success overstates harm by an order of magnitude: the CoSafe attack drives 8B to 73% non-refusal, but only 6% of those responses are actually harmful. Single-turn safety evaluation does not predict how quantized models rank under conversational attack.

1. Introduction

Post-training quantization to 4 bits is now the standard way to fit an instruction-tuned model onto a phone or a laptop. The same deployment target that forces quantization also forces multi-turn use, because a chat assistant holds a conversation. Safety evaluation has not followed. Recent work that measures how quantization changes refusal uses single-turn prompts and reports a drop in refusal after quantization (Kadadekar, 2026). Whether that drop holds, grows, or disappears once an attacker has several turns is open, and the authors name multi-turn as future work.

This paper answers one question: does 4-bit quantization change a model’s multi-turn jailbreak robustness, and how does that compare to its single-turn effect on the same models and prompts? The study covers two models and three quantization methods, and reads two behavioral signals, whether the model refused and whether the response is actually harmful, along with the projection onto the refusal direction of Ardit et al. (2024).

The reported behavioral degradation does not reproduce, at either horizon. On standalone AdvBench prompts, quantization leaves single-turn refusal flat or higher for both models and all three methods. Under a CoSafe coreference attack, precision does not change refusal either, and all precisions converge to a common non-refusal rate (Figure 1a). A Llama Guard 3 rescore confirms the null on a second metric: quantization never raises the rate of actually-harmful responses (Figure 1b). The only significant behavioral precision effect anywhere is an increase, where 8B quantization raises refusal on the CoSafe cold turn, and even that vanishes under the attack.

The practical consequence is a warning about evaluation. A single-turn safety score does not preserve how FP16 and quantized checkpoints rank under conversational attack, and on these models it does not surface a behavioral safety cost from quantization at all. The contributions are:

- A well-powered non-replication across two Llama models, three 4-bit methods, two prompt sets (AdvBench and CoSafe), and two metrics (substring refusal and a Llama Guard harm judge).
- The first measurement of 4-bit quantization under a multi-turn attack. Precision changes neither multi-turn refusal nor multi-turn harm, and the one significant single-turn effect does not survive the attack.
- Two evaluation cautions. Refusal-based attack success overstates actual harm by an order of magnitude on CoSafe, and a safety-prompt positive control shows the harm metric can detect a real intervention.

2. Background

2.1 Quantization and safety

Kadadekar (2026) reports that INT4 quantization degrades single-turn refusal while the refusal direction stays geometrically intact, and notes that standard probes do not explain the behavioral gap. Chhabra and Khalili (2025) find that the refusal direction survives compression with cosine similarity above 0.99. Both study single-turn behavior. Neither measures a conversation. This paper extends the first to the multi-turn setting it names as open.

Llama-3.1-8B under the CoSafe multi-turn attack

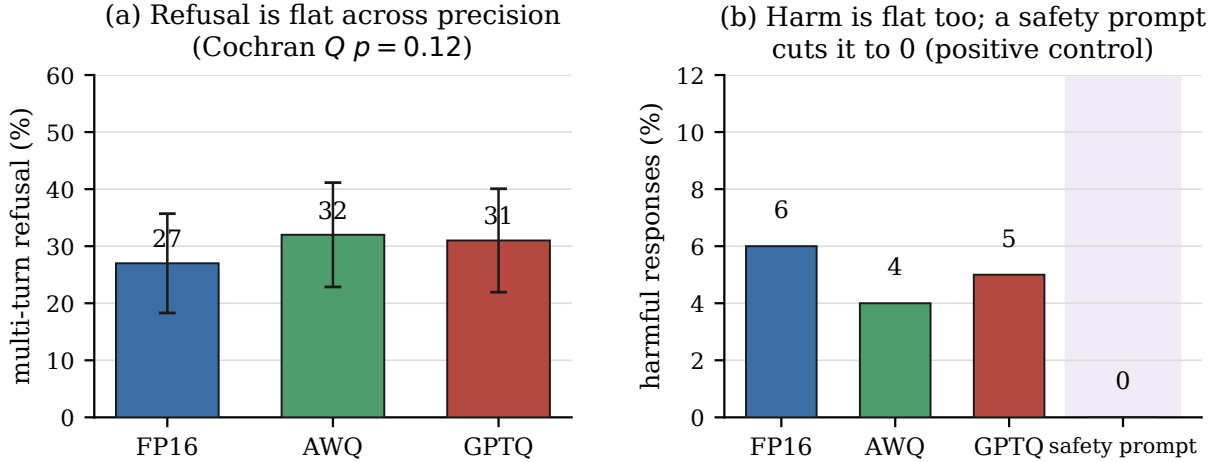


Figure 1: The two-metric null in one view. (a) Multi-turn refusal does not vary with precision (Wilson 95% intervals overlap). (b) Multi-turn harmful-response rate is low and does not rise under quantization, yet a safety system prompt drops it to 0%, showing the metric can detect a real intervention.

2.2 Refusal as a direction

Arditi et al. (2024) show that refusal in instruction-tuned models is mediated by a single linear direction in the residual stream. Adding it induces refusal, and ablating it removes refusal. This study uses their difference-in-means estimator and confirms the causal layer by ablation rather than by separability.

2.3 Multi-turn attacks

STAR (2026) shows that in full-precision models the projection onto the refusal direction decays across conversation turns, so multi-turn state erodes refusal. CoSafe (Yu et al., 2024) builds multi-turn attacks whose final turn refers back to earlier turns by coreference, so the harmful request is only legible in context. The measurement here is independent of STAR and is used to compare precisions.

3. Experimental Setup

3.1 Models and precisions

The 3B model is Llama-3.2-3B-Instruct at FP16 and NF4 (bitsandbytes double-quantized). The 8B model is Llama-3.1-8B-Instruct at FP16, AWQ-INT4, and GPTQ-INT4, using the public hugging-quants checkpoints. AWQ and GPTQ layers are not fused, so forward hooks see the residual stream.

3.2 Refusal direction and causal layer

The refusal direction is extracted by difference-in-means: the mean activation at the post-instruction token for harmful prompts minus the same for harmless prompts, per layer, normalized. Held-out separation accuracy is

100% for the 3B direction. The functional layer is chosen by ablation, not by separability. Each candidate layer’s direction is removed from the residual stream, and the layer whose removal collapses refusal is kept. For 3B this is layer 20 and for 8B layer 12, where ablation drops refusal from 94% to 0% while a random-direction control stays at baseline. The separability-argmax layer differs in both models, 28 for 3B and 31 for 8B, and does not mediate refusal, so only ablation-confirmed layers are used.

3.3 Attacks and prompts

Multi-turn attacks come from CoSafe, using its safe categories (non-violent unethical behavior, financial crime, privacy violation, and misinformation), 100 scenarios in total. For each scenario the final unsafe turn is measured two ways. Cold presents the final turn alone. In-context presents the full conversation, which is the attack. The two conditions share the same final request, so their difference isolates the effect of the conversational context. Standalone single-turn prompts come from AdvBench, 100 instructions presented as one turn, to test the quantization effect on overtly harmful prompts in the same setting as prior single-turn work.

3.4 Metrics

Refusal is detected by a fixed phrase classifier on greedy generation. To measure whether a response is actually harmful, every completion is rescored with Llama Guard 3 (Inan et al., 2023), reporting the harmful-response rate. The projection of the post-instruction activation onto the refusal direction is recorded as a mechanism

Model	Prec.	Refusal		Proj.	
		cold	in-ctx	cold	in-ctx
3B	FP16	25%	23%	+1.46	-0.86
3B	NF4	23%	23%	+1.33	-0.84
8B	FP16	23%	27%	+0.88	+0.82
8B	AWQ	31%	32%	+0.80	+0.76
8B	GPTQ	34%	31%	+0.90	+0.89

Table 1: Refusal and mean projection onto the refusal direction, cold (single turn) and in-context (multi-turn), $n = 100$ per cell.

Comparison	Cold	In-context
3B FP16 vs NF4	$p = 0.68$	$p = 0.62$
8B FP16 vs AWQ	$p = 0.043$	$p = 0.13$
8B FP16 vs GPTQ	$p = 0.010$	$p = 0.22$
8B Cochran Q	$p = 0.008$	$p = 0.12$

Table 2: Refusal significance. The single-turn precision effect on 8B is significant, and the multi-turn effect is not.

readout. Refusal and harm are paired binary across precision, so comparisons use McNemar’s test per pair and Cochran’s Q across the three 8B precisions. Projection is paired continuous, so it uses the paired t -test. Every run is 100 items, so $n = 100$ per condition.

4. Results and Analysis

4.1 Multi-turn refusal does not depend on precision

Table 1 gives the rates. For 3B the two precisions are identical at 23% in-context refusal (McNemar $\chi^2 = 0.25$, $p = 0.62$), and the paired projection difference is null ($t = 0.57$, $p = 0.57$). For 8B the three precisions span 27 to 32%, and Cochran’s Q over the paired refusals is not significant ($Q = 4.2$, $p = 0.12$; Figure 1a). The non-refusal rate sits near 70% for every precision, though most of that is not harmful output.

4.2 The single-turn gap does not survive the attack

The cold column tells a different story for 8B. Quantization raises single-turn refusal, with FP16 at 23%, AWQ at 31%, and GPTQ at 34%. Both pairwise contrasts against FP16 are significant (McNemar $p = 0.043$ and $p = 0.010$; Table 2), and Cochran’s Q is significant ($Q = 9.7$, $p = 0.008$). The same three precisions are not separable once the conversation is present. The 8 to 11 point spread at a single turn narrows to 5 points in context and loses significance (Figure 2). The 3B model shows no precision effect at either horizon.

4.3 No degradation on standalone prompts

On AdvBench single-turn prompts (Table 3) quantization does not lower refusal either. The 3B model moves from 76% to 82% under NF4 (McNemar $p = 0.15$). The 8B model moves from 95% to 98% under AWQ ($p = 0.25$) and stays at 95% under GPTQ ($p = 0.62$).

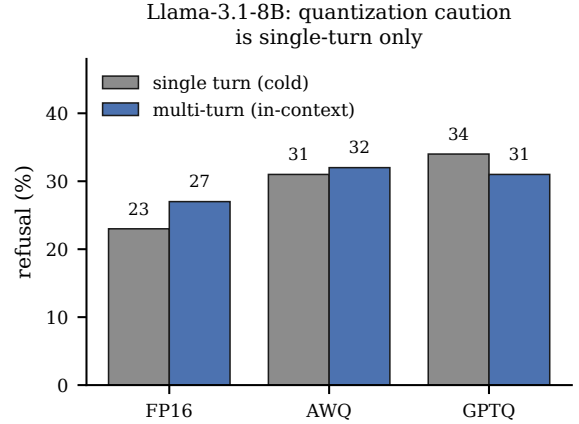


Figure 2: Llama-3.1-8B refusal, single turn (cold) versus multi-turn (in-context). The single-turn precision gap narrows and loses significance in context.

Model	Refusal		McNemar vs FP16
	FP16	quantized	
3B	76%	NF4 82%	$p = 0.15$
8B	95%	AWQ 98% GPTQ 95%	0.25/0.62

Table 3: Single-turn refusal on 100 standalone AdvBench prompts. No precision comparison is significant.

No comparison is significant, and every point estimate is flat or higher under quantization. This is a well-powered non-replication of the single-turn refusal degradation reported for INT4 quantization. Refusal here is near ceiling for 8B, so the harm judge addresses the question from the other side.

4.4 The null holds under a harm judge

Rescoring every completion with Llama Guard 3, which labels whether the response itself is harmful, gives the rates in Table 4. Quantization does not raise the harmful-response rate in any condition. Every quantized run is at or below its FP16 baseline, and no comparison is significant (McNemar $p \geq 0.48$). The null survives a second, stricter metric (Figure 3).

4.5 The harm metric has power: a positive control

A null on a rare-event metric invites the objection that nothing could show up. To rule that out, a safety system prompt is prepended to every FP16 condition, an intervention that should change behavior. It does, and the harm metric detects it. Under the multi-turn attack, the system prompt cuts 8B harmful responses from 6% to 0% (McNemar $p = 0.041$) and raises multi-turn refusal from 27% to 38% ($p = 0.006$; Figure 1b). The harm metric is not saturated. It registers a real safety intervention on the same prompts where quantization registers nothing. The quantization null is a measured absence of effect, not an absence of power.

Quantization moves neither metric, single-turn or multi-turn

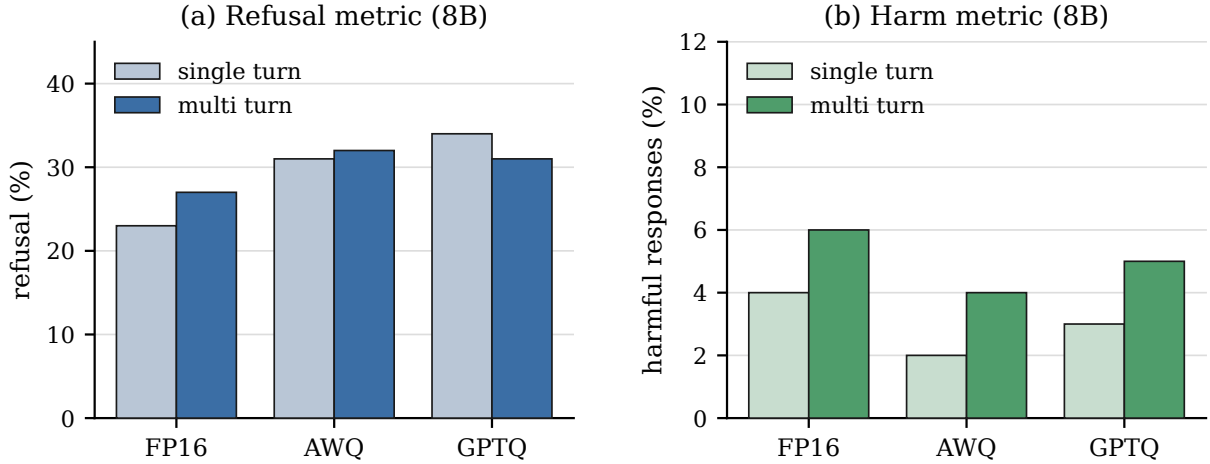


Figure 3: Single-turn versus multi-turn on both metrics for Llama-3.1-8B. Quantization moves neither refusal (a) nor harmful-response rate (b) at either horizon.

Setting	Model	FP16	quantized
single	3B	1%	NF4 1%
single	8B	4%	AWQ 2% / GPTQ 3%
multi	3B	4%	NF4 3%
multi	8B	6%	AWQ 4% / GPTQ 5%

Table 4: Harmful-response rate (Llama Guard 3), lower is safer. Quantization never raises it; all comparisons non-significant.

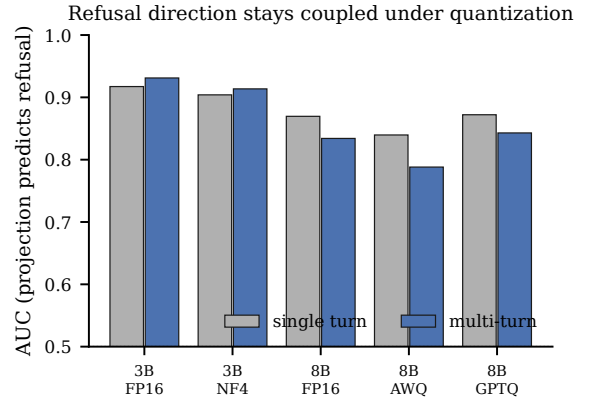


Figure 4: AUC of projection onto the refusal direction predicting refusal. The direction stays coupled to behavior under every precision.

4.6 Refusal-based attack success overstates harm

The two metrics diverge sharply. The CoSafe attack takes 8B to a 73% non-refusal rate by the substring metric, but only 6% of those responses are judged harmful by Llama Guard. The attack mostly produces non-refusing but non-harmful text. On these prompts, refusal-based attack success overstates actual harm by an order of magnitude, which is a caution for how multi-turn safety is reported.

4.7 The direction stays coupled

If quantization broke the refusal representation, the projection would stop predicting refusal. It does not. The AUC of projection predicting refusal ranges from 0.79 to 0.93 across all precisions and conditions (Figure 4), and point-biserial correlations run from 0.46 to 0.68 with $p < 10^{-6}$. Multi-turn convergence is behavioral saturation across a shared threshold, not a loss of the mechanism.

5. Discussion

5.1 Why the multi-turn gap closes

CoSafe attacks hide the harmful request behind coreference, so the model answers the earlier benign-looking turns and then completes the final one in context. The attack is strong, taking FP16 from 23% to 27% refusal for 8B and driving 3B to 23%. Once the attack is this effective, a few points of extra caution from quantization is not enough to change the outcome. A safety margin that exists in isolation does not imply a safety margin under pressure.

5.2 Implication for evaluation

Single-turn refusal is the standard axis for comparing a quantized checkpoint to its FP16 source. This result shows that axis does not preserve the multi-turn ordering. For 8B, single-turn evaluation ranks the quantized checkpoints as safer than FP16, while under attack they are statistically tied. A single-turn number does not transfer to the conversational setting where the model is deployed.

5.3 Relation to prior reports

Kadadekar (2026) reports single-turn degradation on standard harmful prompts. This study does not reproduce it. Across both prompt sets and both metrics the behavioral effect of quantization is null-to-positive, not degradation. The Llama Guard rescore closes the ceiling concern that would otherwise bound a refusal-only null, because on a metric with room below, and one the positive control shows can detect a real intervention, quantization still moves nothing.

6. Limitations and Future Work

The study covers two model sizes (3B and 8B), one family (Llama), one multi-turn attack (CoSafe coreference), and three quantization methods. It does not vary attack type or study training-time quantization. The refusal detector is a phrase classifier, though the harm judge and the projection readout, which do not use it, agree with the behavioral results. Both metrics sit near an extreme, with substring refusal near ceiling and the Llama Guard harm rate near floor, so each alone has limited room to expose a small effect. The positive control mitigates this for the harm metric, but the low harm rate partly reflects the CoSafe safe categories being mild relative to the severe-hazard taxonomy Llama Guard is tuned for. The result is best read as: on these prompts, actual harmful output is rare at every precision, not as a guarantee on severe-harm categories. A stricter judge, a second multi-turn attack, and a larger model family are the direct next steps.

7. Conclusion

Across two Llama models, three 4-bit methods, two prompt distributions, and two metrics, 4-bit quantization does not reduce safety at either a single turn or across a multi-turn attack. The only significant behavioral precision effect is an increase, where 8B refuses more on the CoSafe cold turn, and it does not survive the attack, where all precisions converge near 30% refusal and 4 to 6% harmful output. A safety system prompt, by contrast, does lower multi-turn harm on the same prompts, so the null is a true absence of effect rather than a lack of measurement power. Single-turn safety evaluation does not predict how quantized models rank under conversational attack.

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